



IDBI INNOVATE

2026

Build. Integrate. Transform.

Knowledge Partner



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Team Details

- a. **Team name:** Black Bulls
- b. **Team leader name:** Shubham
- c. **Problem Statement:** Open Track — AI copilot for channel-sourced retail lending (conversational AI · product advisory · financial-health scoring · default prediction)

Brief about the idea

A channel-aware loan-origination & partner-relationship copilot, **on top of IDBI's existing LOS / BRE / LMS.**

Retail lending is sourced through DSAs & dealers and runs on TAT + trust — so it works all four sides at once.

CUSTOMER

Only suitable, clearly-explained, instant offers — no spam, no black box.

RM / BANK

A queue ranked by risk-adjusted profit, with a reason for every score.

CHANNEL PARTNER (DSA)

Instant decisions, transparent auto-payouts, tiering — the loyalty moat.

RISK / REGULATOR

Explainable, fair, adverse-selection-aware lending with an audit trail.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

HOW IT'S DIFFERENT

- **Channel-first.** Models the DSA relationship — adverse-selection, TAT, tiering, churn.
- **Bank-grade economics.** Ranks by $P(\text{convert}) \times (\text{net interest income} - \text{expected credit loss})$.
- **Explainable.** Per-decision reason codes + Reg-B / RBI adverse-action codes.
- **Integrates, not replaces.** An intelligence layer on top of LOS / BRE / LMS.

HOW IT SOLVES THE PROBLEM

- **Speed + fit.** Right lead → right product → right price, instantly, with a pro offer.
- **Lower NPAs.** Flags risky sourcing before disbursal, not after.
- **Partner loyalty.** Transparent, fast, tiered payouts → more & better volume.
- **The flywheel.** Win the partner → volume → data → sharper models.

List of features offered by the solution

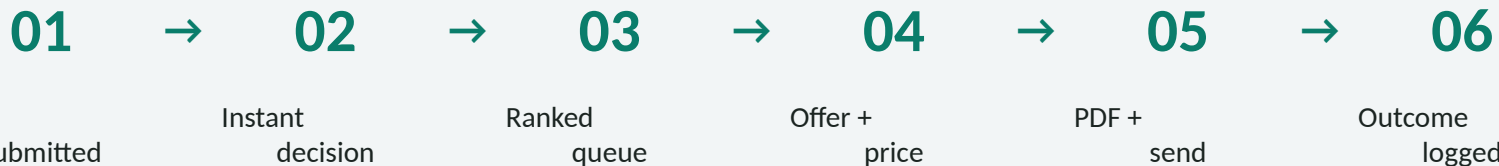
CUSTOMER & RM

- ML conversion-propensity queue (risk-adjusted)
- Explainable financial-health score (0–100)
- Deterministic eligibility, rate & EMI engine
- Suitability-first next-best-product advisory
- Editable offer + risk-based pricing in-band
- Premium branded PDF + WhatsApp / Gmail send
- Per-decision reason codes on every score

CHANNEL & PARTNER

- Partner / lead default-risk (adverse-selection)
- Channel-mix ROI + book concentration (HHI)
- Commission + clawback payout engine
- Duplicate / fraud lead detection
- Instant decision (<50 ms) + SLA / TAT clock
- Partner tiering + churn early-warning
- Agentic AI assistant (natural language)

Process flow diagram or Use-case diagram

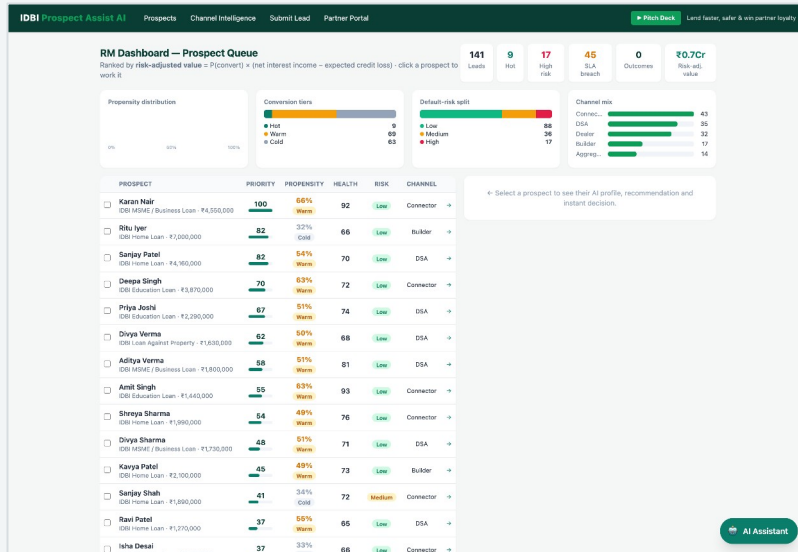


🔄 **Closed loop.** Outcomes (won / lost / default) retrain the ML models — the compounding data moat.

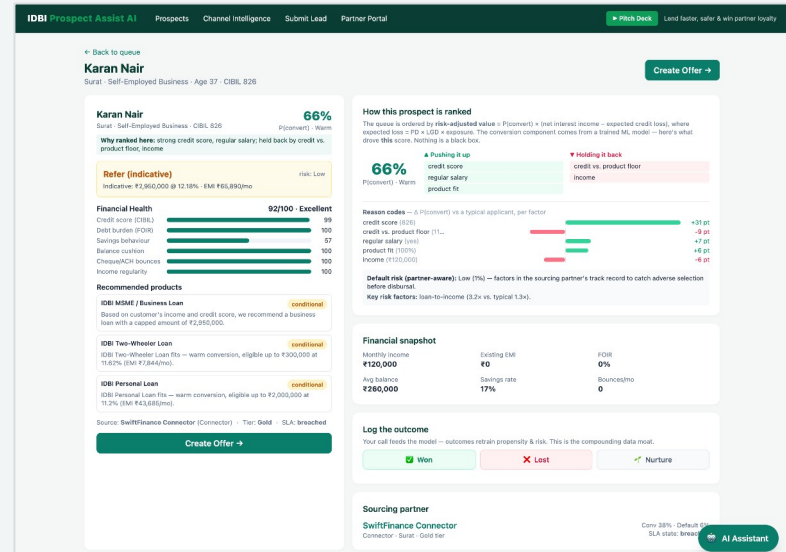
Numbers boundary. The deterministic engine owns every rupee (eligibility, rate, EMI, ECL); the ML owns probabilities; the LLM only phrases. The RM stays in control — every edit re-validates through the engine, so no invalid or non-compliant offer can ever leave.

Wireframes/Mock diagrams of the proposed solution (optional)

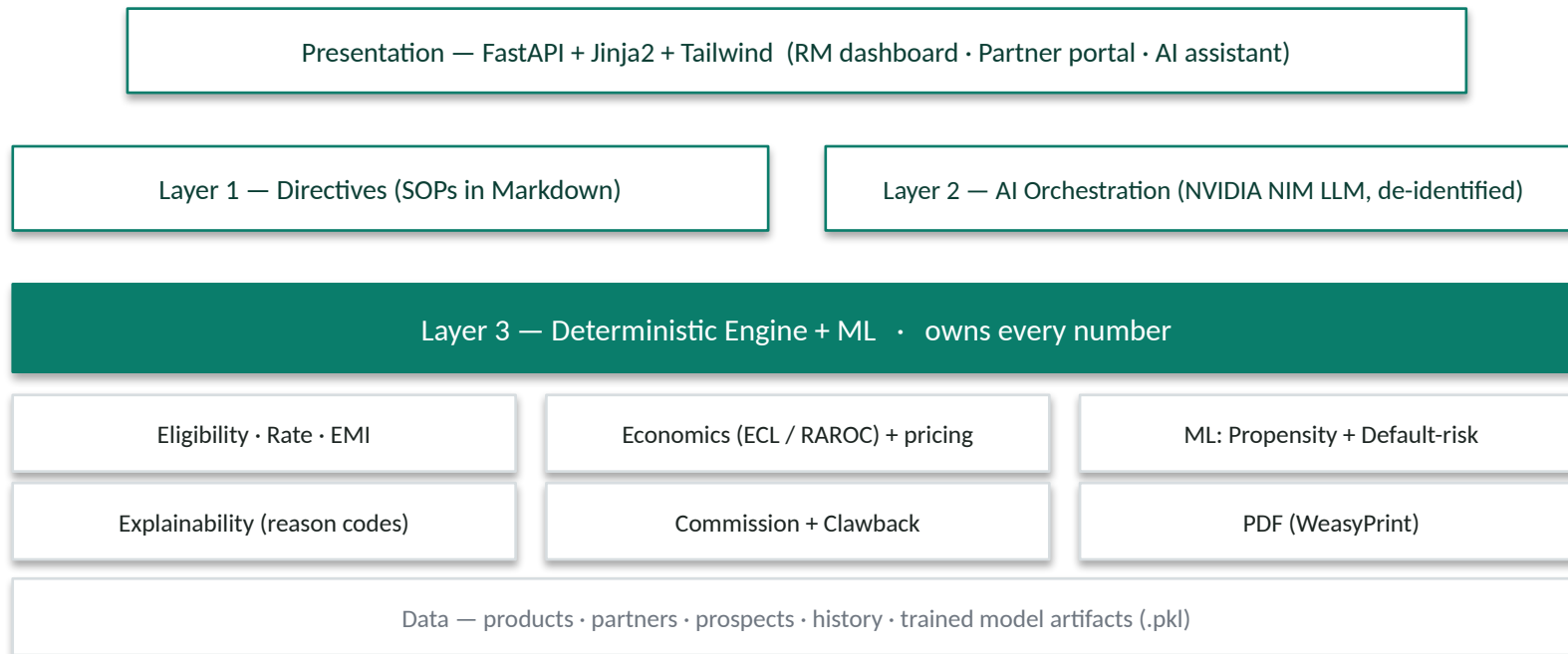
RM DASHBOARD — RISK-ADJUSTED QUEUE



PROSPECT PROFILE — REASON CODES



Architecture diagram of the proposed solution



Technologies to be used in the solution

BACKEND & WEB

Python 3.14 • FastAPI • Uvicorn • Jinja2 • Tailwind CSS • vanilla JS

MACHINE LEARNING & AI

scikit-learn • NumPy / SciPy • joblib • occlusion XAI • NVIDIA NIM LLM

DOCUMENTS & PDF

WeasyPrint • pango / cairo

QUALITY & DEPLOYMENT

pytest (44 tests) • Playwright • Docker • Render / Railway / Cloud Run

Estimated implementation cost (optional)

PROTOTYPE — TODAY

≈ ₹0

Open-source stack

NVIDIA NIM free tier

Free-tier hosting · no licence cost

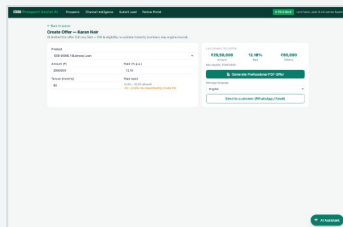
PRODUCTION POC — INDICATIVE (ANNUAL)

- LLM / GPU inference — usage-based
- Cloud hosting + DB + storage — modest
- Integration — LOS / BRE / KYC / CIBIL (one-time)
- Model-ops — monitoring, retraining, governance

ROI — cost saved (NPA reduction + payout-leakage → 0 + RM selling hours) >> cost to run.

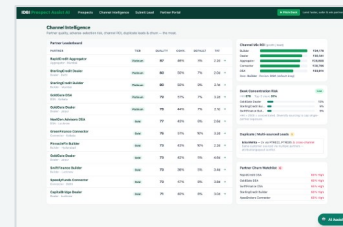
Snapshots of the prototype

EDITABLE OFFER — LIVE EMI & RISK-BASED PRICING



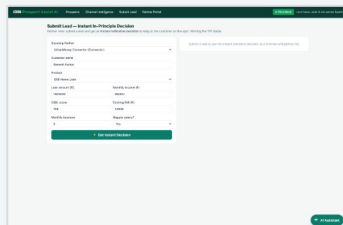
The screenshot shows a form titled 'Editable Offer - Basic Info'. It includes fields for 'Loan Amount', 'Tenure (in months)', and 'EMI'. There are also dropdowns for 'Loan Type' and 'Risk Profile'. A 'Calculate EMI' button is visible at the bottom right of the form.

CHANNEL INTELLIGENCE — ADVERSE SELECTION, ROI, HHI



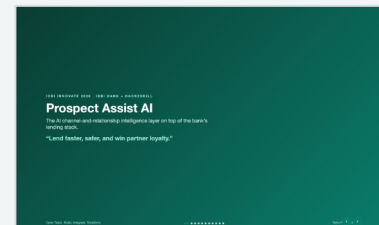
The screenshot shows a dashboard titled 'Channel Intelligence'. It features a table with columns for 'Channel', 'Adverse Selection', 'ROI', and 'HHI'. The table lists various channels like 'Direct', 'Agent', 'Referral', etc. To the right of the table is a bar chart showing the distribution of these metrics.

INSTANT DECISION + ADVERSE-ACTION REASON CODES



The screenshot shows a form titled 'Instant Decision - Instant Decision'. It includes fields for 'Loan Amount', 'Tenure (in months)', and 'Decision Status'. There is a 'Get Decision' button at the bottom right of the form.

BUILT-IN PITCH DECK (LIVE NUMBERS)



Prototype Performance report/Benchmarking

MODEL PERFORMANCE — HOLD-OUT, SYNTHETIC (INDICATIVE)

0.69 / 39

CONVERSION PROPENSITY — AUC / GINI

0.81 / 62

PARTNER DEFAULT-RISK — AUC / GINI

SPEED & INTEGRITY

< 50 ms

INSTANT DECISION LATENCY

~1-2 s

LLM REPLY · 9 S TIMEOUT

0.22

FEATURE SHARE — NO LEAKAGE

BUSINESS IMPACT — SYNTHETIC BOOK

₹16.4 Cr

PIPELINE VALUE

₹0.68 Cr

RISK-ADJUSTED VALUE

44 / 44

AUTOMATED TESTS PASSING

Additional Details/Future Development (if any)

01 Integrate

- IDBI sandbox APIs
- CIBIL bureau + KYC
- e-consent (DPDP Act)
- LOS / BRE / LMS hooks

02 Harden

- Retrain on real conv / NPA / TAT
- SHAP + PSI drift monitoring
- Auth / RBAC + audit trail
- Data residency

03 Scale

- Partner mobile portal + WhatsApp
- OTP-gated secure delivery
- Collections & Early-Warning
- NVIDIA AI Enterprise / on-prem

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GITHUB PUBLIC REPOSITORY

<https://github.com/Shubham-33/Hack2Skill-IDBI-Hackathone>

FINAL PRODUCT LINK (LIVE PROTOTYPE)

<https://prospect-assist-ai-2guf.onrender.com>

DEMO VIDEO LINK (3 MINUTES)

[paste your unlisted YouTube / Drive link]



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Thank you!